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Andreja Drolc · Magda Cotman · Milenko Roš

Integration of metrological principles in a proficiency-testing scheme in the field of water analysis

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Abstract Measurement traceability is universally recognised as one of the basic prerequisites for comparability of results obtained in different laboratories and is a basic aspect of metrological sciences such as analytical chemistry. This requirement is underscored by the increasing adoption of standards and measurement quality systems, such as laboratory accreditation against ISO/IEC 17025. Testing laboratories ensure traceability of their measurement results by using appropriate reference standards for calibration of instruments and control of measurement processes. For routine work in the field of water analysis, these standards are usually commercial solutions or in-house solutions prepared from "pure" products. Therefore, laboratories should demonstrate that their use of reference standards is appropriate and sufficient, which can be done by participation in an appropriate proficiency-testing scheme. The paper reports how measurement traceability of results from field laboratories (nitrite nitrogen, nitrate nitrogen, chloride and sulphate; all in water) can be demonstrated by participation in a proficiency-testing scheme based on reference values.

Keywords Wastewater analysis · Traceability · Proficiency testing · Reference value

Introduction

The monitoring and protection of the environment is based on measurements. Industrial emissions—potential sources of environmental contamination—are measured in order to protect the environment. Only reliable analytical results can serve as a basis for drawing meaningful conclusions about the situation and the evolution of existing or potential contaminants [1].

To achieve comparability of results between different laboratories, it is essential to link all the individual measurement results to some common, stable reference or measurement standard; the results can therefore be compared through their relationship to that reference. Metrological traceability is universally recognised as one of the basic prerequisites for comparability of the results from different laboratories and is a basic aspect of any metrological science such as analytical chemistry. Traceability is defined as "the property of the result of a measurement or the value of the standard whereby it can be related to stated references, usually national or international standards, through an unbroken chain of comparisons, all having stated uncertainties" [2]. Pure substance certified reference materials (CRMs) are often used for appropriate calibration of comparative instrumental methods. A CRM is defined as a "reference material, accompanied by a certificate, one or more of whose property values are certified by a procedure which establishes traceability to an accurate realization of the unit in which the property values are expressed, and for which each certified value is accompanied by an uncertainty at a stated level of confidence" [2]. The CRMs are normally characterised by a primary method, which is higher in the metrological chain than a field method.

These requirements are underscored by the increasing adoption of standards and measurement quality systems, such as laboratory accreditation against ISO/IEC 17025 [3], which is widely adopted in the environmental sector. Testing laboratories ensure traceability of their measurement results by using appropriate reference standards for calibration of their instruments [4]. In real-life routine work, traceability of the results from testing laboratories in the field of water analysis is usually not established using CRMs, produced and certified by national metrological institutes. In most cases these

A. Drolc (✉) · M. Cotman · M. Roš
National Institute of Chemistry,
Hajdrihova 19, P.O. Box 660, 1000 Ljubljana, Slovenia
E-mail: andreja.drolc@ki.si
Tel.: +386-1-4760238
Fax: +386-1-4760300

standards are commercial solutions or in-house solutions prepared from “pure” products. However, laboratories should demonstrate that their use of reference standards is appropriate and sufficient. This can be done by participation in a proficiency test (PT), which integrates metrological principles. A PT is a type of interlaboratory comparison where the objective is to obtain an indication of the performance of the laboratory. The performance of the laboratory is measured against the assigned value. In general, PTs can be operated in two different ways. Usually, PTs are based on consensus values, which depend on the results of the participants and serve as a basis to assess the performance of the laboratories. Another approach is to work with a reference value, which is obtained independently and is not connected to the results of the participants. When metrological traceability of the reference value is established using CRMs, then participation in such a PT scheme enables participants to demonstrate the traceability of their results. On the other hand, a PT scheme based on a consensus value contains no information about traceability of results.

The purpose of the present paper is to illustrate how measurement traceability of results from field laboratories can be demonstrated by participation in a proficiency-testing scheme using reference values. A detailed description of a PT in the field of water analysis is given: preparation of the test material, the results of homogeneity and stability testing, reference measurements for establishing the reference value, and setting performance criteria for the participants. The results from the PT scheme operating with reference values were compared to results from a PT scheme operating with consensus values.

General description of the proficiency-testing scheme

In Slovenia, there are about 50 laboratories performing chemical water analysis. In order to asses the quality of field testing laboratories, a PT scheme was organised by the National Institute of Chemistry (NIC), Ljubljana, Slovenia, in collaboration with the Metrology Institute of the Republic of Slovenia (MIRS). The scheme involved four measurands of chemical water analysis (chloride, sulphate, nitrite nitrogen and nitrate nitrogen in water). The Laboratory for Water Chemistry, Biology and Technology at the National Institute of Chemistry is accredited according to standard ISO/IEC 17025, certified according to ISO 9001, and is holder of the national reference etalon, based on the authorisation and cofounding of the Metrological Institute of the Republic of Slovenia [5–8].

The PT scheme, which was carried out in accordance with ISO Guide 43-1 [9] and ILAC requirements [10], was operated with a reference value in order to demonstrate the traceability of results from the participating laboratories.

Test material preparation, homogeneity, stability

Proficiency-testing material (water samples) was prepared by the organiser. All characteristics that could affect the integrity of the test were considered, including the homogeneity and stability of the samples. Samples for chloride, sulphate, nitrite nitrogen and nitrate nitrogen were prepared at two different concentration levels for each analyte (Table 1). Where possible, the test material was similar to samples that are analysed in the participating laboratories so that analytical errors could be detected in the methods applied. This means similarity of matrix composition, the contents of the analytes, the mode of binding of these analytes, pattern of possible interferences and the physical status of the material. In some cases, analytes in the wastewater matrix were not stable for the duration of the PT study, so they were prepared in deionised water. The test material, prepared in a wastewater matrix, was taken from the inflow to a municipal wastewater treatment plant. The material was collected and prepared in an amount sufficient to supply all participants with one litre of each test sample. To avoid inhomogeneity of the samples, the wastewater was filtered through a special filter device (pore size 0.45 μm) to remove particulate matter. Where necessary, some samples were spiked with an additional amount of analyte to adjust the concentration level. Samples were stored at 4°C and delivered to participants in polyethylene containers.

The inhomogeneity of the material should not significantly affect the total uncertainty of the measurement. The homogeneity of all the samples was tested by analysing ten randomly-selected samples. All measurements were performed in triplicate. The homogeneity testing results were used to estimate the between-bottle homogeneity. All of the prepared samples fulfilled the prescribed criterion: that the between-bottle standard deviation should be lower than 1/3 of the standard deviation of the results from all participants [11].

After having verified the homogeneity, the stability of the samples was studied. The material has to be stable in order to ensure that identical samples are delivered to all participating laboratories. The study was performed to verify that the integrity of the material is maintained under the severest conditions that may be encountered.

Table 1 Characteristics of the test material

Sample identification	Matrix	Analyte	Concentration level, mg/kg
A1	Deionised water	Cl^-	0.5–50
		SO_4^{2-}	0.5–50
A2	Municipal wastewater	Cl^-	50–500
		SO_4^{2-}	50–500
A1N	Deionised water	NO_2^- –N	0.5–5
		NO_3^- –N	0.5–5
A2N	Deionised water	NO_2^- –N	5–50
		NO_3^- –N	5–50

Preliminary studies revealed that the main problem was connected to the stability of chloride and sulphate in wastewater samples at low concentration levels (0.5–50 mg/kg) and the general stability of wastewater samples containing nitrite nitrogen and nitrate nitrogen. The instabilities of the samples might be dictated by biological, physical or chemical phenomena, closely connected to the matrices of the samples. Only wastewater samples containing chloride and sulphate at high concentration levels (50–500 mg/kg) were stable at 4 °C for the required period. Other samples were stable only when the analytes were prepared in deionised water (samples A1, A1N and A2N, refer to Table 1). Therefore, to ensure the stability of the samples over the duration of the PT scheme, samples at low level concentrations (0.5–50 mg/kg for sulphate and chloride) and samples containing nitrite nitrogen and nitrate nitrogen were prepared in deionised water by spiking. Data are summarised in Table 1. The stabilities of all the samples at 4 °C were confirmed by trend analysis for the period of duration of the PT study (three weeks) [12].

Reference measurements

An optimised procedure for obtaining an acceptable estimate of the measurand was established, including calculation and a set of measurement conditions. The method was validated to demonstrate that the calculation and the conditions prescribed are sufficiently complete for the purpose. After that, traceability or control for each value in the equation and for each of the specified conditions was established. Traceability was established by calibration using appropriate measurement standards. All of the data required were obtained from calibration certificates, method validation data and quality control data. Reference standards were chosen to be appropriate to the equipment being calibrated, of appropriate uncertainty, and their values were traceable to relevant references. The measurement uncertainty of the procedure was evaluated.

Measurement procedure

The complete procedure for anions (chloride, nitrite, nitrate and sulphate) in wastewater is stated in the International Standard ISO: "Water quality—determination of dissolved anions by liquid chromatography of ions—part 2: determination of bromide, chloride, nitrate, nitrite, orthophosphate and sulphate in wastewater" [13]. Anions are separated using liquid chromatographic separation by means of a separating column (Dionex 120, injection volume 100 μl). An anion exchanger is used as the stationary phase (guard column AG4A-SC, separation column AS4A-SC) and aqueous solutions of salts of weak monobasic and dibasic acids as mobile phases (1.8 mM carbonate/1.7 mM bicarbonate, flow of eluent 2.0 ml/min). Conductivity detection is used in this method (background conductivity 15–20 μS). The conductivity detector is combined with a suppressor device (AMMS-2), which decreases the conductivity of the eluent and converts the separated anions into their corresponding acids. Determination of chloride, nitrite, nitrate and sulphate ion is performed by calibration of the overall procedure. The calibration curve is established by measuring a series of reference solutions (approximate concentrations: 0.5, 1, 2, 3, 4, 5, 6 and 7 mg/kg).

It was evident from experimental data that the peak area-value errors increased as the analyte concentration increased, indicating heteroscedastic data. The regression line is calculated to give additional weight to those points where the errors are the smallest. Therefore, the concentration of each anion, C, is calculated using a weighted regression method:

$$C = \frac{A - a_{\mathrm{w}}}{b_{\mathrm{w}}} \tag{1}$$

where A is the measured area of the sample chromatographic peak, b_{w} the slope of the weighted linear calibration curve and a_{w} the intercept. The weights w_{i}, the slope of the linear weighed calibration curve b_{w} and the weighted intercept a_{w} are calculated from Eqs. 2–6:

$$w_i = \frac{n}{s_i^2 \sum_i 1/s_i^2} \tag{2}$$

$$b_{\mathrm{w}} = \frac{\sum_i w_i \cdot C_i \cdot A_i - n \cdot \overline{C}_{\mathrm{w}} \cdot \overline{A}_{\mathrm{w}}}{\sum_i w_i \cdot C_i^2 - n \cdot \overline{C}_{\mathrm{w}}^2} \tag{3}$$

$$a_{\mathrm{w}} = \overline{A}_{\mathrm{w}} - b_{\mathrm{w}} \cdot \overline{C}_{\mathrm{w}} \tag{4}$$

$$\overline{C}_{\mathrm{w}} = \sum_i w_i \cdot C_i / n \tag{5}$$

$$\overline{A}_{\mathrm{w}} = \sum_i w_i \cdot A_i / n \tag{6}$$

where C_{i} is the concentration of the reference solution at the ith level (C_1,... C_{i},... C_{n}), A_{i} is the area of the chromatographic peak of the ith reference solution (A_1,... A_{i}, ... A_{n}), s_{i}^2 is the corresponding variance, and n is the total number of calibration reference solutions.

Method validation

The method is an established standardised method, which has already been validated. Therefore, only selected parameters were considered to verify the correct implementation of the method in the laboratory [14]: trueness and precision. A reference material was analysed (RTC, Anions, I-051, Lot P2) and no significant discrepancy between our results and the certified value

Table 2 Analysis of reference material RTC I-051, Lot P2—determination of accuracy

Measurand	Certified value (mg/kg)	Analysed value 1 (mg/kg)	Analysed value 2 (mg/kg)	Analysed value 3 (mg/kg)
Cl^-	24.7 (22.9–26.5)	24.9 ± 0.4 ($k = 2$)	24.7 ± 0.4 ($k = 2$)	24.7 ± 0.4 ($k = 2$)
SO_4^{2-}	34.5 (28.2–40.8)	34.4 ± 0.5 ($k = 2$)	34.5 ± 0.5 ($k = 2$)	34.5 ± 0.5 ($k = 2$)
NO_2^- –N	1.50 (1.28–7.73)	1.51 ± 0.02 ($k = 2$)	1.52 ± 0.02 ($k = 2$)	1.50 ± 0.02 ($k = 2$)
NO_3^- –N	14.8 (11.8–17.8)	14.8 ± 0.2 ($k = 2$)	14.7 ± 0.2 ($k = 2$)	14.7 ± 0.2 ($k = 2$)

Table 3 Calibration RMs used

Measurand	Producer	Identification	Clasification	Concentration	Measurement uncertainty
Chloride	NIST	SRM 3182	CRM	997 mg/kg	0.7 mg/kg ($k = 2$)
Nitrite	Ultra scientific	ICC-007	RM	1000 mg/l	0.2% ($k = 2$)
Nitrate	NIST	SRM 3185	CRM	976 mg/kg	6 mg/kg ($k = 2$)
Sulphate	NIST	SRM 3181	CRM	1001 mg/kg	4 mg/kg ($k = 2$)

was found (Table 2). The laboratory also participates in PT (AQUACHECK, UK) and good performance in this determination has been obtained. The most common measurements of precision are repeatability and reproducibility, which were also determined in the laboratory.

Establishing traceability

Pure substance reference materials were used to calibrate the instrument in order to establish measurement traceability [15–18]. Data are summarised in Table 3. For samples with a wastewater matrix, the method of standard addition was also used to avoid matrix effects in the measurement results [19].

Traceability of mass was provided by regular calibration procedures for the balance used, and they were confirmed by the associated calibration certificate from an accredited laboratory (traceable to DKD-K-21201). The validity was checked on a daily basis with internal check weights, which are traceable to national standards.

Measurement uncertainty evaluation

The sources of uncertainty [20, 21] in the determination of anions by ion chromatography are schematically presented in the cause and effects diagram shown in Fig. 1.

Uncertainty components were quantified in one of the following ways: experimentally from a series of repeated observations by calculating the standard deviation (type A evaluation), or obtained from other sources such as information from calibration certificates (type B evaluation). Before calculating the combined uncertainty, type B uncertainties were expressed

Fig. 1 The sources of uncertainty in chloride, nitrite, nitrate and sulphate determination—cause and effects diagram

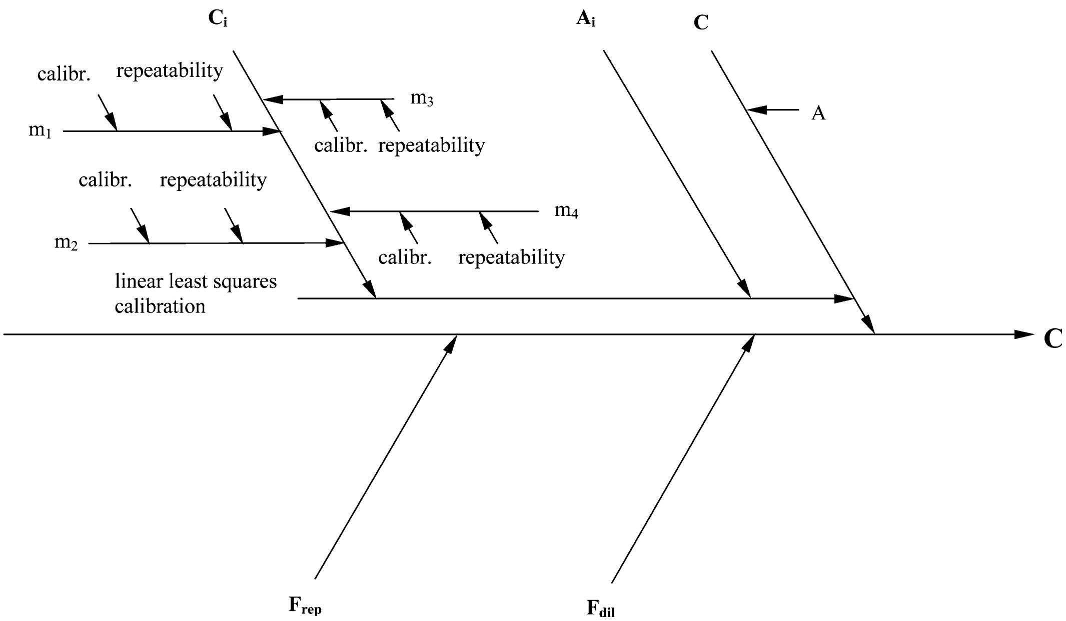


Table 4 Uncertainty components of C_{NO_3-N}(sample identification A1N) and their standard uncertainties

Quantity	Symbol	Unit	Value	Standard uncertainty
Concentration of solution 1	C_1	mg/kg	0.488	0.0019
Concentration of solution 2	C_2	mg/kg	0.976	0.0035
Concentration of solution 3	C_3	mg/kg	1.952	0.0070
Concentration of solution 4	C_4	mg/kg	2.928	0.0110
Concentration of solution 5	C_5	mg/kg	3.904	0.0140
Concentration of solution 6	C_6	mg/kg	4.880	0.0190
Concentration of solution 7	C_7	mg/kg	5.856	0.0210
Concentration of solution 8	C_8	mg/kg	6.832	0.0250
Area of the reference solution 1	A_1	–	51437	76
Area of reference solution 2	A_2	–	98583	204
Area of reference solution 3	A_3	–	200037	151
Area of reference solution 4	A_4	–	305822	312
Area of reference solution 5	A_5	–	418793	315
Area of reference solution 6	A_6	–	531318	712
Area of reference solution 7	A_7	–	645226	716
Area of reference solution 8	A_8	–	767727	295
Area of sample	A	–	496187	–
Dilution factor	F_{dil}	–	1.980	0.004
Repeatability	F_{rep}		1.000	0.007
Factor N/NO_3	F	–	0.225	–
Nitrate nitrogen concentration	$C_{rm\ NO}3{-}N$	mg/kg	2.02	0.015[a]

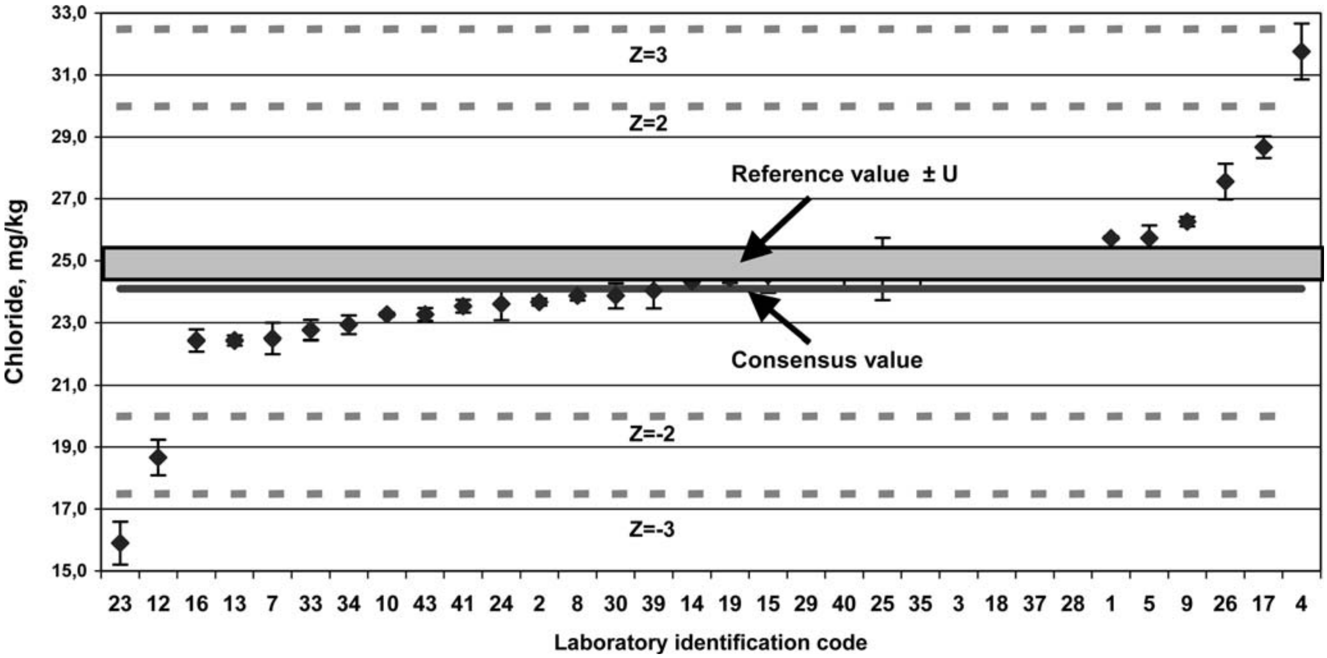
[a] Calculated according to Eq. 7

as one standard deviation. If there were no data on the type of distribution, it was estimated as being rectangular or triangular, and then converted to a normal distribution by dividing by factors of $\sqrt{3}$ or $\sqrt{6}$, respectively.

Evaluating uncertainty components

Uncertainty components (see Fig. 1, Table 4) are associated with the following input quantities: uncertainty associated with linear least square calibration; overall repeatability of the procedure. The uncertainty associated with the weighted regression method was evaluated as follows. The amount of each anion was calculated using a previously-prepared weighted calibration curve. The uncertainty of the calculated concentration obtained from the calibration curve is associated with the uncertainty of the calibration solution concentrations and the uncertainty of the measured peak areas of the reference solutions. Diluted working reference solutions were prepared by transferring an aliquot of the CRM to an empty, dry, pre-weighted bottle and than reweighing the bottle. From these data the exact concentrations of the working solutions were calculated. Dilutions prepared gravimetrically need no correction for temperature and no further correction for true concentration in

Fig. 2 Laboratory performance in the PT scheme (determination of chloride, sample A1–Cl)



a vacuum. The calibration certificate of the balance identifies two uncertainty sources for tared weighing: the calibration function of the scale, and the repeatability. The calibration function has two potential uncertainty sources, identified as the sensitivity of the balance and its linearity. The sensitivity was neglected because the mass was weighed by the same balance over a very narrow range. The linearity was obtained from the manufacturer's certificate, where it was reported with a 95% confidence limit and converted to one standard deviation by dividing by a factor of 2, while the repeatability of weighing was obtained from successive weighing operations based on data from control charts. Because two successive weighings are used, uncertainties have to be counted twice. The uncertainties were then combined by taking the square root of the sum of the squares of the individual components. Uncertainties in peak areas are mainly associated with variations in the flow of the mobile phase. All repeatability contributions were combined into one contribution for the overall measurement procedure and obtained from the method validation data.

Calculation of the combined and the expanded uncertainties

When the result y is calculated from input quantities x_1, x_2,... x_n, than the uncertainty of the result is calculated following the law of propagation of errors:

$$u(y) = \sqrt{\sum_{i=1}^{n} \left(\frac{\partial y}{\partial x_i}\right)^2 \cdot u(x_i)^2} \tag{7}$$

where $u(x_i)$ are the standard uncertainties of the input parameters, and $\partial y / \partial x_i$ is a sensitivity coefficient. The sensitivity coefficient describes how the measurement result varies with changes in the values of input estimates. Equation 7 is valid for measurements where there is no correlation between the input parameters.

As an example, the uncertainty sources for nitrate nitrogen determination are presented. The values of the input parameters C_1, C_2, C_3, C_4, C_5, C_6, C_7, C_8, A_1, A_2, A_3, A_4, A_5, A_6, A_7, A_8, A, F_{rep} and F_{dil} of nitrate determination, as well as their respective standard uncertainties, are given in Table 3. The result is corrected by the factor N/NO_3 to express the result as N. To calculate the expanded uncertainty of the result of a measurement at the 95% confidence level, the result for the combined uncertainty was multiplied by a coverage factor of 2.

Reference values for the proficiency-testing scheme, obtained via reference measurements, are listed in Table 5.

Evaluation of the results of the proficiency-testing scheme

Participants were requested to analyse samples by their usual techniques and to make three separate determinations of each measurand. Laboratories with procedures for measurement uncertainty evaluation were asked to also report measurement uncertainty. Out of 254 results, approximately one third were accompanied with measurement uncertainty statements. The results from the participants were analysed using the statistical procedure of ISO 5725-2 [22]. Results were evaluated on the basis of assigned values, which were obtained in two fundamentally different ways. Reference values were determined on the basis of reference measurements performed at the National Institute of Chemistry. In addition, a consensus value, based on the results from participants, was also calculated in order to compare the results of a PT based on a reference value and those based on a consensus value. The consensus value was calculated from the results from participants after elimination of outliers. Cochran's and Grubb's tests were used to eliminate the highest value in a set of standard deviations and the largest and the smallest observations, respectively. The consensus values, accompanied by the reproducibility standard deviation between laboratories, are summarised in Table 5. The reference values, obtained on the basis of reference measurements and the consensus values, which are based on the results from participants, are in very good agreement, showing that the consensus values are not biased.

Table 5 Determination of the assigned values—reference values and consensus values

Sample identification	Parameter	Reference value		Consensus value		
		Result (mg/kg)	Measurement uncertainty (mg/kg) ($k = 2$)	Result (mg/kg)	Number of results	Reproducibility standard deviation (mg/kg)
A1	Cl^-	25.0	0.5	24.1	28	2.17
A2	Cl^-	208	4	210	24	6.22
A1N	NO_2^- –N	4.07	0.07	4.10	25	0.21
A2N	NO_2^- –N	15.6	0.3	15.6	27	0.64
A1N	NO_3^- –N	2.02	0.03	2.13	25	0.29
A2N	NO_3^- –N	24.3	0.4	25.0	26	2.24
A1	SO_4^{2-}	37.5	0.5	34.7	31	2.53
A2	SO_4^{2-}	266	2	265	33	23.2

Table 6 Performance criteria (CV_{perf}) for result evaluation

Sample identification	Concentration level	CV_{perf}
A1 (Cl^-, SO_4^{2-})	0.5–50 mg/kg	0.1 (10%)
A2 (Cl^-, SO_4^{2-})	50–500 mg/kg	0.05 (5%)
A1N (NO_2^-–N, NO_3^-–N)	0.5–5 mg/kg	0.1 (10%)
A2N (NO_2^-–N, NO_3^-–N)	5–50 mg/kg	0.05 (5%)

The internationally recommended Z-score was used as a performance criteria. For each result X_i, the individual Z-score was calculated according to Eqs. 8 and 9:

$$Z = \frac{X_i - \text{assigned value}}{s_{perf}} \tag{8}$$

$$s_{perf} = CV_{perf} \cdot \text{assigned value} \tag{9}$$

where the assigned value is either the reference or the consensus value and CV_{perf} is the target performance criterion. The target performance criteria were known by participants in advance and were determined according to analytical method performance and the concentration level of the analyte. Target performance criteria for different measurands are listed in Table 6.

When interpreting Z-scores, the following assessments were made:

- $|z| \leq 2$, the result is satisfactory
- $2 < |z| < 3$, the result is questionable
- $|z| \geq 3$, the result is unsatisfactory.

Table 7 Performances of the laboratories in the PT study

Lab. Code	Z-Score															
	A1–Cl		A2–Cl		A1N–NO_2		A2N–NO_2		A1N–NO_3		A2N–NO_3		A1–SO_4		A2–SO_4	
	Ref.	Cons.	Ref.	Cons.	Ref.	Cons.	Ref.	Cons.	Ref.	Cons.	Ref.	Cons.	Ref.	Cons.	Ref.	Cons.
1	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	$2<z<3$	$2<z<3$
2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	$2<z<3$	< 2	< 2	< 2	< 2	< 2
3	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2
4	$2<z<3$	> 3	< 2	< 2	< 2	< 2	< 2	< 2	> 3	> 3	> 3	> 3	< 2	< 2	< 2	< 2
5	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2
6					< 2	< 2	< 2	< 2								
7	< 2	< 2	< 2	< 2									< 2	$2<z<3$	< 2	< 2
8	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2
9	< 2	< 2	> 3	> 3	< 2	< 2	< 2	< 2	> 3	$2<z<3$	< 2	< 2	< 2	< 2	< 2	< 2
10	< 2	< 2	< 2	< 2					> 3	> 3	> 3	> 3				
11					< 2	< 2	$2<z<3$	< 2	> 3	$2<z<3$	< 2	$2<z<3$				
12	$2<z<3$	$2<z<3$	> 3	> 3												
13	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2
14	< 2	< 2	< 2	< 2					< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2
15	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2
16	< 2	< 2	< 2	< 2	< 2	< 2	$2<z<3$	$2<z<3$	< 2	< 2	$2<z<3$	> 3	< 2	< 2	< 2	< 2
17	< 2	< 2	< 2	< 2									$2<z<3$	$2<z<3$	< 2	< 2
18	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2
19	< 2	< 2	> 3	> 3	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2
21					< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2				
22					< 2	< 2	< 2	< 2	> 3	> 3	> 3	> 3				
23	> 3	> 3	> 3	> 3	< 2	< 2	< 2	< 2	> 3	> 3	> 3	> 3				
24	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2
25	< 2	< 2	< 2	< 2	< 2	< 2	$2<z<3$	$2<z<3$	< 2	< 2	$2<z<3$	> 3				
26	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	> 3	> 3	> 3	> 3	< 2	< 2	< 2	< 2
27					< 2	< 2	< 2	< 2	> 3	> 3	> 3	> 3				
28	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2
29	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2
30	< 2	< 2	< 2	< 2									< 2	< 2	< 2	< 2
31													< 2	< 2	> 3	> 3
32					< 2	< 2	< 2	< 2	$2<z<3$	$2<z<3$	< 2	$2<z<3$	< 2	< 2	$2<z<3$	$2<z<3$
33	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	> 3	> 3
34	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	$2<z<3$	< 2	< 2	< 2	< 2	< 2	< 2	< 2
35	< 2	< 2	< 2	< 2			< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2
36													> 3	> 3	> 3	> 3
37	< 2	< 2	< 2	< 2									< 2	< 2	< 2	< 2
38													< 2	< 2	> 3	> 3
39	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2
40	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2
41	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2			< 2	< 2
42													< 2	< 2	< 2	< 2
43	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2

Ref. Z-score is calculated according to reference value, *Cons.* Z-score is calculated according to consensus value

As an example, results from the PT scheme for determining chloride in sample A1–Cl, based on reference values, are presented in Fig. 2. The results for the performances of the laboratories, obtained via the reference values, showed that out of 254 results reported, 24 were unsatisfactory (9.4%) and 13 were questionable (5.1%) (Table 7). This shows that the majority of Slovenian laboratories analysing water operate under appropriate traceability conditions. The laboratories with unsatisfactory performance used either standardised or in-house methods using commercial kits. There were no general rules that performance might depend on the method used. Laboratories with unsatisfactory performance should re-examine their procedures for traceability establishment and revalidate their methods. The results show that there is no significant difference between the performances of the laboratories when the assigned value was denoted as a consensus or a reference value. Out of a total of 254 results, only ten of them (3.9%) depended on the approach used. For all of the measurands, the consensus values and the reference values were in very good agreement.

Conclusions

In order to demonstrate the traceability of results from Slovenian laboratories operating in the field of water analysis, a proficiency-testing scheme was organised by the National Institute of Chemistry, Ljubljana, Slovenia in collaboration with the Metrological Institute of the Republic of Slovenia. Selected measurands included chloride, sulphate, nitrite nitrogen and nitrate nitrogen.

PT material with a wastewater matrix was found to be unstable during the duration of the study for all of the measurands. The instabilities of the samples might be dictated by physical, chemical or biological phenomena. Therefore, to ensure that identical samples were delivered to all participating laboratories, spiked samples had to be used in some cases.

The assigned values for measurands, included in the proficiency-testing scheme, were obtained on the basis of reference measurements. Traceability or control for each value in the equation and for each of the specified conditions was established. Finally, the measurement uncertainty of the procedure was evaluated. The performances of the laboratories were evaluated by calculating their Z-scores. Results from the proficiency testing, based on reference values, indicated that only 9.4% of the results (from a total of 254 reported results) were unsatisfactory and 5.1% were questionable. This suggests that the majority of the Slovenian laboratories analysing water operate under appropriate traceability conditions. Laboratories participating in PT, which includes metrological principles, are able to demonstrate traceability to SI units and reliable analytical quality.

A comparison between the performances of the laboratories, using PT schemes based upon either reference value or consensus value (based on the results from all participants), was also made. Results from the comparison revealed that there was no significant difference between the reference values and consensus values for any of the measurands. Consequently, the difference in performance scores (Z-scores) calculated using reference values or consensus values was negligible.

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